DEPI | Digital Egypt Pioneers Initiative

OnboardX

AI-Powered Automated Client Onboarding System

Case study: FlexiPay (Consumer Finance and BNPL)

Project Documentation - Parts 1 to 3

Digital Egypt Pioneers Initiative - Graduation Project

Item	Details
Project name	OnboardX
Developing team (brand)	OnboardX Automation Systems
Case-study client	FlexiPay - consumer finance and BNPL (Buy Now Pay Later)
DEPI track	AI Automation Workflow With N8N Professional
Team / group	CLS ONL5_AIS9_S3
Team leader	Shenouda Romany
GitHub repository	Eng-Shenouda/Final-Project
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1. Project Planning and Management

1.1 Project Proposal

Overview. OnboardX is an AI-powered client onboarding system that lets a consumer finance company activate a new customer in minutes, at the moment of purchase, instead of days. The customer scans a QR code or opens the app, uploads a national ID and takes a selfie. The system then reads the ID, verifies that the person and the document are genuine, screens the person against AML and PEP lists, scores the risk, and returns an approval with a credit limit. Only doubtful cases are sent to a human reviewer. Every step is written to a protected audit log.

Case-study client. The project is built around **FlexiPay**, a consumer finance and BNPL company that lets individuals get a fast credit line and pay for goods and services in instalments at online stores and physical shops without complicated paperwork. **OnboardX Automation Systems** is the team (the solution provider) that designs and builds the onboarding system for it.

The five operational problems

#	Problem at FlexiPay	Business impact	How OnboardXaddresses it
P1	**High drop-off at the point of sale.** Customers leave the store or website when they learn that application means manual forms and a wait of two days to a week.	Lost sales for merchants, lost customers for FlexiPay	Automated activation target of about 3 minutes: QR or app, ID upload, selfie, instant decision and credit limit while the customer is still in the store
P2	**Identity fraud and fake accounts.** Forged national IDs or stolen identities are used to buy on instalments without paying.	Direct financial losses	[Future Enhancement] Planned architecture: face matching, liveness detection, and document authenticity checks (scheduled for full production rollout phase).
P3	**Slow manual data entry and errors.** Staff or customers type name, national ID number and address by hand.	Typos, delayed credit enquiries, rework	AI OCR engine reads the national ID or passport and fills the system fields automatically; target accuracy above 98%
P4	**High operating cost and workload.** Large teams manually review thousands of daily instalment requests.	Rising expenses and bottlenecks	Smart risk scoring and decision engine: target of about 80% of clean, compliant requests approved automatically; about 20% flagged for human review
P5	**Complex KYC and AML compliance.** Manually checking each customer against anti-money-laundering and sanctions lists is hard.	Regulatory exposure, no reliable record	[Future Enhancement] Planned compliance module: automated AML and PEP list screening with audit logging (designed as a scalable integration for production phase).

> **About the numbers.** The 3-minute activation time, the 98% OCR accuracy and the 80/20 automatic-versus-review split come from the project brief. In this document they are treated as **design targets**, and the testing phase will measure what the prototype actually achieves on test data. No real customer data will be used (see Section 1.4, risk R3 and Section 3.5).

Proposed solution at a glance

Step	What happens	AI or automation component
1. Start	Customer scans a merchant QR code or opens the app and gives consent for data processing	Intake and QR service
2.Capture	Customer photographs the national ID (or passport) and takes a selfie; image quality is checked and a retake is requested if needed	Capture guidance
3. Read	ID fields (name, national ID number, address, date of birth) are extracted and validated automatically	AI OCR engine
4. Verify	Face on the ID is matched with the selfie; liveness and document authenticity are checked	Face matching, liveness detection, document authenticity
5. Screen	Customer data is screened against sanctions, PEP and blocked-person lists	AML and PEP screening
6. Decide	Risk score is computed; the engine approves and assigns a credit limit, or flags the case for review	Smart risk scoring and decision engine
7. Notify	Customer and merchant receive the result at once; flagged cases appear on the reviewer dashboard	Notifications and review dashboard
8. Record	Every input, check result, decision and human action is logged with time and actor	Audit log

Project objectives

- **O1 - Speed:** complete onboarding in about 3 minutes from QR scan to decision for a clean application (addresses P1).
- **O2 - Fraud resistance:** detect forged documents, face mismatches and non-live captures before any credit is granted (P2).
- **O3 - Accuracy:** extract ID data automatically with more than 98% field accuracy and remove manual typing (P3).
- **O4 - Efficiency:** approve about 80% of eligible applications automatically and route about 20% to human reviewers with the reasons attached (P4).
- **O5 - Compliance and traceability:** screen every applicant and keep a tamper-resistant record of every decision (P5).

Project scope

In scope (prototype)	Out of scope
QR and app-style web intake flow with consent capture ID and passport capture with quality checks AI OCR extraction and field validation (including national ID number structure) Face matching, liveness detection and document authenticity check AML and PEP screening against list data Risk scoring, decision engine and credit limit rule Reviewer dashboard for flagged cases (approve, reject, request more information) Notifications to customer and merchant Audit log and basic reporting	Real credit bureau connection (the prototype uses a mock enquiry) Real payment processing and instalment collection Production-grade fraud and credit models trained on real FlexiPay data Use of real customers' ID images or live production deployment Loan servicing, collections and debt recovery Native iOS/Android store releases

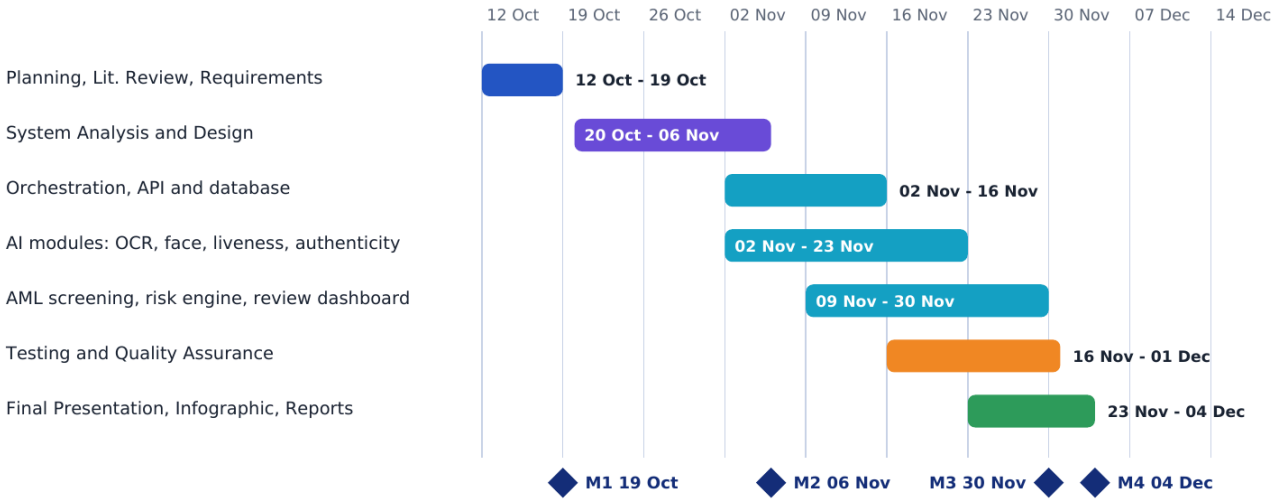
Key assumptions: FlexiPay is a case-study client, so business data, volumes and thresholds are assumptions that a real deployment would replace; test data is synthetic or fictional; the final choice of AI tooling (open-source models or vendor APIs) is made in the design phase.

1.2 Project Plan

The plan follows the deadlines in the Project Documentation Guidelines (DEPI column). Parts 1 to 3 are submitted on 19 October 2026, as announced by CLS Learning Solutions.

Phase	Period	Main deliverables	Milestone
Planning, Literature Review, Requirements	12 - 19 Oct	This document (Parts 1-3), GitHub repository created	M1 - 19 Oct
System Analysis and Design	20 Oct - 6 Nov	Use case, DFD, sequence, activity, state and class diagrams; ERD and schema; architecture; wireframes; AI tooling decision	M2 - 6 Nov
Implementation	2 - 30 Nov	Working onboarding flow end to end: intake, OCR, identity verification, AML screening, risk engine, dashboard, audit log; README and deployment link	M3 - 30 Nov
Testing and QA	16 Nov - 1 Dec	Test plan and cases, accuracy and speed measurements, security checks, bug reports	-
Final Presentation and Reports	23 Nov - 4 Dec	User manual, technical documentation, presentation, infographic, optional demo video	M4 - 4 Dec

Gantt chart



Work packages

WP	Work package	Description	Lead role	Window
WP1	Intake and QR	Entry page, QR link, consent, capture UI, retake loop	Frontend	Nov 2 - 16
WP2	OCR engine	Extract and validate ID and passport fields, accuracy tests	AI engineer	Nov 2 - 23
WP3	Identity verification	Face match, liveness, document authenticity, scoring of results	AI engineer	Nov 2 - 23
WP4	AML and PEP screening	List ingestion, name matching with Arabic and English variants, hit handling	Backend	Nov 9 - 25
WP5	Risk and decision engine	Scorecard or model, credit limit rule, reason codes, thresholds	AI engineer	Nov 9 - 30
WP6	Orchestration and API	Workflow that connects all modules, retries, error handling, notifications	Backend	Nov 2 - 16
WP7	Review dashboard and audit log	Flagged-case queue, decisions, evidence view, log and reports	Frontend	Nov 9 - 30
WP8	Quality, security and compliance	Test plan, privacy and security checks, bias and accuracy evaluation	QA lead	Nov 16 - Dec 1

Resource allocation

Tool choices below are candidates; the final selection is recorded in the design documents.

Resource	Purpose	Note
Workflow orchestration (n8n planned)	Connects intake, AI services, screening, decision and notifications	Open-source; self-hosted or trial
OCR engine	Reads Arabic and English text from ID images	Open-source library or cloud OCR API
Face matching and liveness	Compare ID photo with selfie; detect non-live captures	Open-source models or vendor API; simplified challenge-based liveness acceptable for the prototype
Document authenticity checks	Detect tampering, edited fields, screen or photocopy capture	Image-forensics rules plus model; limited by lack of real forged samples
Backend API and database	Applicants, documents, checks, decisions, audit log	Final stack documented in the technology stack section of the design phase
Screening lists	Sanctions, PEP and blocked-person data for matching	Public or sample lists for the prototype
Test data	Synthetic or fictional ID images and applicants	No real personal data
GitHub and team time	Version control, documentation, collaboration	About 2-3 per member over 8 weeks

1.3 Task Assignment and Roles

Roles follow the work packages. One person may hold more than one role; replace the highlighted names with the team members.

Role	Responsibilities	Work packages	Member
Team Leader and Project Manager	Plan, track milestones, coordinate with the Technical Instructor, own the GitHub repository and the final submission	All (coordination)	Shenouda Romany
Business and Requirements Analyst	Stakeholder analysis, requirements, use cases, literature review, traceability	Docs, WP8 support	Malak Khaled
AI and Data Engineer	OCR, face matching, liveness, document authenticity, risk model, test datasets	WP2, WP3, WP5	Basmla Talal
Backend and Workflow Engineer	Orchestration, API, database, AML screening, notifications, audit log	WP4, WP6	Ahmed Elsayed
Frontend and UX Engineer	Customer capture flow, reviewer dashboard, wireframes and UI guidelines	WP1, WP7	Shenouda Romany
QA, Security and Compliance Lead	Test plan and cases, bug reports, privacy and security review, accuracy and bias evaluation	WP8	Maryam Hany
Documentation and Presentation Lead	User manual, technical documentation, infographic, final presentation, demo video	Final phase	All

Working agreements: all documents and code go to the single team GitHub repository with meaningful commit messages; every change is reviewed by a second member; the team meets weekly to compare progress with the Gantt chart.

1.4 Risk Assessment and Mitigation Plan

L = likelihood, I = impact (Low, Medium, High).

ID	Risk	L	I	Mitigation
R1	Very short timeline for an AI system with many modules	High	High	Fix the prototype scope (Section 1.1); build the happy path end to end first, then add checks; weekly checkpoint against the plan
R2	OCR accuracy drops on blurry, glared or angled photos and on Arabic text	High	Medium	Capture guidance and quality checks with retake loop; confidence scores per field; low-confidence fields go to review; measure accuracy on a varied test set
R3	No access to real IDs, real fraud samples or real FlexiPay data	High	High	Use synthetic or fictional data; state clearly that results are indicative; design thresholds as configurable
R4	Face matching or liveness gives false rejections or can be spoofed	Medium	High	Tune thresholds with a test set; route borderline results to review instead of auto-reject; test against photo and screen replay attacks
R5	Risk model is biased or unexplainable	Medium	High	Start with a transparent scorecard and reason codes; evaluate results across test groups; human review for borderline cases and every rejection
R6	Biometric and financial data are sensitive under Egyptian data protection law [14]	Medium	High	Collect explicit consent, minimize and delete data when the purpose ends, restrict access, encrypt, mask national IDs in logs
R7	Cloud AI APIs may move personal data outside Egypt, which the law restricts without a licence [14]	Medium	High	Prefer locally hosted models; use only synthetic data with any external API during the prototype
R8	AML name matching misses matches or creates false hits (Arabic transliteration, name variants)	Medium	High	Fuzzy matching with name variants; any possible hit goes to a compliance reviewer; log every screening result

ID	Risk	L	I	Mitigation
R9	Over-reliance on automation: wrong automatic approvals	Medium	High	Conservative thresholds; human review of flagged cases; credit limit caps; periodic review of decisions
R10	Third-party service downtime, latency or cost	Medium	Medium	Timeouts and retries, fallback to manual review, alert on failure, cost limits on API usage
R11	Scope creep (adding scoring features, payments, apps)	Medium	Medium	Out-of-scope list agreed by the team; new ideas go to a future-enhancements backlog
R12	Unequal workload or unavailable team member	Medium	Medium	Backup owner per work package, shared documentation, cross-review

1.5 Key Performance Indicators (KPIs)

Targets are goals for the final demo. Where a baseline is needed (for example drop-off), it will be estimated and recorded during the implementation phase.

Area	KPI	How it is measured	Target
Speed	Activation time for a clean application	Time from QR scan to decision on test runs (median and 90th percentile)	About 3 minutes median
Speed	System processing time (excluding customer capture)	Time from last upload to decision	Under 30 seconds
Accuracy	OCR field-level accuracy	Correct fields / total fields on a labelled test set	Above 98%
Fraud control	Forged or tampered test documents detected	Detected / seeded forgeries in the test set	At least 90% (to be refined)
Fraud control	Spoof attempts (photo, screen replay) rejected by liveness	Rejected / attempted spoofs	At least 90% (to be refined)
Identity	False rejection of genuine applicants by face matching	Genuine pairs rejected / genuine pairs	Below 5% (to be refined)
Automation	Straight-through approval rate	Auto-approved / eligible clean applications	About 80%
Automation	Share routed to human review	Flagged / total applications	About 20%
Compliance	Applicants screened against AML and PEP lists	Screened / total applicants	100%
Compliance	Decisions with a complete audit record	Decisions with log entry / total decisions	100%
Reliability	Availability during test period	Uptime of the onboarding service	99% or higher
Business	Reduction in manual effort per application	Reviewer minutes per application before versus after	At least 70% (to be measured)
Usability	Customers completing capture without help	Successful self-service completions in user tests	90% or higher

2. Literature Review

This review covers: (a) evidence on why onboarding friction and identity fraud matter for BNPL, (b) what technologies and solutions exist and how they compare with VeriFlow, and (c) the Egyptian legal and regulatory context that shapes the design. Bracketed numbers refer to the source list in Section 2.7.

2.1 Why onboarding speed and identity assurance matter

- **BNPL is growing fast and attracts fraud.** ACI reported that BNPL grew 237% in the first half of 2024 compared with the same period in 2023, as cited by Trulioo [3]. Rapid growth with instant approval creates an attractive target for fraudsters.
- **Document steps are where customers leave.** A benchmarking report by the Payments Innovation Forum with HooYu found that on average only 74% of customers complete onboarding because of KYC abandonment, and half of the fintechs surveyed see their biggest drop-out when customers must provide ID documents [1].
- **Waiting increases abandonment.** HooYu's research reports that the longer a prospective customer waits for KYC to finish, the higher the abandonment rate [2]. This matches FlexiPay's problem P1: the customer is in the shop and will not wait days.
- **Automation works best with people in the loop.** Vendor articles citing an ABBYY survey report that companies combining human interaction with automation improved customer experience and retention more than pure automation [4]. This supports VeriFlow's design of automatic decisions for clean cases and human review for doubtful ones.

> **Reading note:** [2], [4] and the vendor pages [5], [6], [11] promote their own products, so their figures and claims are treated as indicative, not as independent proof.

2.2 Fraud types in BNPL and the recommended controls

Trulioo lists four main BNPL fraud types: synthetic identity fraud (real and fake data combined), merchant fraud (fake merchant accounts), account takeover, and new-account fraud where a stolen or false identity opens an account and early activity looks normal because the first repayment often comes about a month later [3]. Its recommended defences include document and biometric verification (which can detect presentation attacks and spoofing), risk-based onboarding where checks scale with risk, transaction limits that trigger extra verification, risk signals such as phone and email history, and continuous monitoring [3].

Fraud type [3]	Where it hits FlexiPay	OnboardX control
Synthetic or stolen identity	Forged or borrowed national IDs (P2)	Document authenticity, face match, liveness, national ID number validation
New-account fraud	Fake account opened and instalments never repaid (P2)	Risk scoring, conservative first credit limit, step-up review for flagged cases
Account takeover	Later misuse of an activated account	Future enhancement: step-up authentication; outside the onboarding prototype
Merchant fraud	Fake merchants using the QR flow	Out of scope for the prototype; noted as a future enhancement

2.3 Technology review by component

Component	What the literature and industry sources report	Implication for VeriFlow
ID capture and OCR	Identity vendors advertise automated ID capture and data extraction in about a second or two [6]; document capture and validation is a standard part of digital KYC journeys [7].	Build OCR extraction with confidence scores and validation rules; measure accuracy on Arabic ID images; send low-confidence cases to review
Face matching and liveness	Selfie capture, liveness detection and facial biometrics are standard in digital KYC [7]; liveness is positioned as a defence against synthetic identities and deepfake-assisted onboarding [6].	Combine face match with liveness; treat borderline results as review cases rather than automatic rejections
Document authenticity	Vendors describe AI-driven document analysis at account creation to detect tampered or fake documents [6].	Use image-forensics checks and a model; acknowledge limited forged samples for training in the prototype
Risk-based decisioning	Risk teams can configure verification workflows and step-up checks that vary by merchant, order value and user profile [6]; checks should scale with risk so good customers move quickly [3].	Scorecard with reason codes and configurable thresholds; different limits by risk band
AML and PEP screening	Egyptian AML rules require customer identification, PEP identification (including relatives and close associates), screening against UN Security Council and domestic terrorist lists, and record keeping for at least five years after the relationship ends [8].	Automatic screening at onboarding, human review of any possible hit, retention of screening records and the audit log

2.4 Similar solutions and comparison

Identity and eKYC providers

Solution	What it offers (as reported)	Strengths	Limits for our use case
Digified (Egypt) [9]	FRA approved its AI-powered digital onboarding, eKYC and e-contracting services for non-banking financial institutions in August 2025; digital ID creation (biometric and non-biometric), authentication against national ID and mobile registry data, and digitally signed contracts. The company cites over 25 million verification transactions a year.	Regulator-approved, local, government data authentication	Reviewed source does not describe AML screening, risk scoring or a reviewer dashboard; an outsourced service rather than a flow owned by the lender
HooYu [7]	Digital KYC with selfie capture, liveness detection, ID capture and validation, facial biometrics, address proofing and geolocation; used by Virgin Money for account and credit card journeys.	Mature identity verification; configurable journeys	Focused on the identity step; not tailored to Egyptian IDs or to a BNPL decision flow
Microblink [6]	BNPL-focused identity platform: ID capture and document verification, liveness against deepfakes, step-up authentication, and configurable verification workflows by merchant, order value and user profile.	Closest to the BNPL use case; configurable risk thresholds	Commercial product; page gives no BNPL loss or approval figures; no information on Egyptian ID support

BNPL providers in Egypt

valU and Sympl are Egyptian BNPL players. In January 2023 they partnered so that valU customers could use Sympl's interest-free three-month plans at partner merchants; Sympl targets Egypt's banked segment of about 30 million people, lets users pay in instalments without prior registration, and accepts credit and debit cards [10]. The sources reviewed do not describe their onboarding technology, so this review treats them as evidence of a growing market and of different onboarding models (card-based versus ID-and-credit-line based), not as technical benchmarks.

Capability comparison

Capability	Digified [9]	HooYu [7]	Microblink [6]	VeriFlow
Egyptian national ID and government data checks	Yes	Not reported	Not reported	National ID structure validation; mock data check in prototype
ID OCR and data extraction	Not reported	Yes	Yes	Yes
Face match and liveness	Biometric eKYC	Yes	Liveness (face match not stated)	Yes
Document authenticity check	Not reported	ID validation	Yes	Yes
AML and PEP screening in the same flow	Not reported	Not reported	Not reported	Yes
Risk scoring and automatic decision with credit limit	Not reported	Not reported	Configurable workflows	Yes
Human review dashboard for flagged cases	Not reported	Not reported	Decision center for risk teams	Yes
Full audit log of every step	Not reported	Not reported	Not reported	Yes
Merchant QR in-store activation	Not reported	Not reported	Not reported	Yes

"Not reported" means the reviewed sources do not mention the feature, not that the product lacks it. The comparison is based on public descriptions, not hands-on testing.

2.5 Regulatory and legal context in Egypt

Source of requirement	What it says (summary)	Design implication for VeriFlow
Consumer Finance Law No. 18 of 2020 [12][13]	Governs consumer finance in Egypt; consumer finance companies and providers are supervised by the Financial Regulatory Authority (FRA); covered goods and services include vehicles, durable goods, education, medical services and travel.	OnboardX assists the licensed company's own process; it does not replace its legal duties or approval authority
FRA acceptance of eKYC [9]	FRA approved an AI-powered eKYC and e-contracting provider for non-banking financial institutions in 2025.	Digital onboarding with biometrics is accepted in this sector; a real deployment would still need FRA-aligned procedures
AML Law No. 80 of 2002 and related rules [8]	Customer identification and verification, PEP identification and enhanced monitoring, sanctions screening, and record keeping for at least five years after the relationship ends; financing companies are covered.	Screen every applicant; escalate possible hits to compliance; retain records and the audit log for the required period (to be confirmed with compliance)
Personal Data Protection Law No. 151 of 2020 [14]	Biometric and financial data are sensitive; consent is required; data may be used only for its stated purpose and deleted when it expires; breaches must be reported within 72 hours; transferring data outside Egypt needs a licence; fines apply.	Explicit consent screen, data minimization and retention rules, encryption and access control, in-country processing, breach procedure

This summary is for engineering design only and is not legal advice. The team should confirm obligations with FlexiPay's compliance and legal advisers before any real deployment.

2.6 Lessons applied and how OnboardX differs

- **Be fast where customers drop:** focus on the document and wait-time stages, which the research identifies as the main drop-off points [1][2].
- **Layer checks by risk:** apply stronger verification and lower limits to riskier profiles instead of treating every applicant the same [3][6].

- **Keep humans for the doubtful cases:** automatic approval for clean cases, review for flagged ones, with reasons shown to the reviewer [4].
- **Fill the integration gap:** the solutions reviewed each cover part of the problem (identity checks, e-contracts, BNPL products). OnboardX combines OCR, identity verification, AML and PEP screening, risk decision, human review and audit log in one flow designed around FlexiPay's in-store scenario.

Limitations of this review

- Several sources are vendor pages and press articles; independent academic evaluations of accuracy were not part of this review and can be added in the design phase.
- No Egyptian ID dataset or real fraud data was available, so claims about accuracy will come from the team's own tests on synthetic data.

2.7 Sources

[1] Fintech Times - PIF report finds onboarding failures are costing fintechs billions
thefintechtimes.com/pif-report-finds-onboarding-failures-are-costing-fintechs-billions

[2] HooYu - Preventing onboarding dropout
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[4] Lorikeet - Reducing KYC drop-off with proactive AI conversations
lorikeetcx.ai/articles/reducing-kyc-dropoff-proactive-ai

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[11] Lorikeet - How AI is transforming customer onboarding in fintech
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insightplus.bakermckenzie.com/bm/banking-finance_1/egypt-regulating-consumer-finance

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[14] Ibrachy & Dermarkar - Personal Data Protection Law No. 151 of 2020
ela.law/firms/ibrachy-dermarkar/articles/personal-data-protection-law-no-151-for-year-2020-arab-republic-of-egypt

3. Requirements Gathering

Requirements come from the FlexiPay problem analysis (Section 1.1), the literature review (Section 2) and the project brief. They should be validated with a practitioner from a finance company when one is available.

3.1 Stakeholder Analysis

Stakeholder	Role	Needs and expectations	Influence
Applicant (end customer)	Applies for credit at a shop or online	Fast and simple flow, clear instructions, fair decision, privacy of ID and face data, clear reason and next step if not approved	High
Merchant and cashier	Offers FlexiPay at the point of sale	Instant result while the customer is present, simple QR flow, fewer abandoned sales	High
Review analyst (FlexiPay)	Handles flagged cases	Case queue with evidence (ID image, selfie, scores, reasons), quick approve or reject, no retyping	High
Risk manager (FlexiPay)	Owns credit policy	Configurable thresholds and limits, visibility of approval and fraud rates, explainable decisions	High
Compliance and AML officer	Ensures KYC and AML compliance	Every applicant screened, hits escalated, complete protected audit log, retention rules	High
Operations manager	Owns cost and throughput	Lower manual workload, processing time and volume reports	Medium
IT and administrator	Runs and maintains the system	Monitoring, error alerts, secure credential handling, simple configuration	Medium
Data protection officer	Protects personal data	Consent records, minimization, retention and deletion, breach procedure	High
FRA (external regulator)	Supervises non-banking finance	Practices consistent with consumer finance and AML rules [8][12]	Indirect
Personal Data Protection Center	Supervises data protection	Lawful processing of sensitive and biometric data, controlled cross-border transfer [14]	Indirect

3.2 User Stories and Use Cases

Personas

- **Mona, 27, shopper:** wants to buy a laptop on instalments in a shop today; will leave if the process takes more than a few minutes or needs paperwork.
- **Karim, review analyst:** handles the flagged applications; needs the evidence and reasons in one screen to decide fast and defend the decision later.

User stories

ID	As a ...	I want ...	So that ...	Priority
US-01	applicant	to start by scanning a QR code or opening the app	I do not need forms or a branch visit	Must
US-02	applicant	to be told clearly what data is collected and to give consent	I understand how my data is used	Must
US-03	applicant	guidance and a retake option when my ID photo is unclear	I do not fail because of a bad photo	Must
US-04	applicant	my ID details filled in automatically	I do not type and make mistakes	Must
US-05	applicant	an instant decision with my credit limit	I can buy while I am in the shop	Must
US-06	applicant	a clear message and next step if my case needs review	I know what happens and how long it takes	Must

ID	As a ...	I want ...	So that ...	Priority
US-07	merchant	to see the customer's activation result immediately	I can complete the sale	Must
US-08	review analyst	a queue of flagged cases with ID image, selfie, scores and reasons	I can decide quickly	Must
US-09	review analyst	to approve, reject or ask for more information with a recorded reason	decisions are consistent and traceable	Must
US-10	risk manager	to configure score thresholds and credit limit rules	policy changes without code changes	Should
US-11	compliance officer	automatic AML and PEP screening of every applicant	no one is activated without screening	Must
US-12	compliance officer	possible list matches to be escalated with the matching details	I can investigate and record the outcome	Must
US-13	compliance officer	a tamper-resistant audit log of all steps and decisions	I can prove compliance to regulators	Must
US-14	operations manager	a report of volumes, approval and review rates and processing times	I can track performance and cost	Should
US-15	administrator	alerts when a service fails or times out	I can fix problems before customers notice	Must
US-16	data protection officer	records of consent and automatic deletion after the retention period	the company stays within the data protection law	Should

Use cases

	UC-01 Instant onboarding at the shop	UC-02 Review a flagged case	UC-03 Handle a possible AML or PEP hit
Actors	Applicant, merchant, system	Review analyst, system	Compliance officer, system
Precondition	Merchant has an active FlexiPay QR code	Application flagged by the decision engine	Screening returned a possible match
Main flow	1. Applicant scans QR and gives consent. 2. Applicant photographs the ID and takes a selfie. 3. System reads the ID and checks quality. 4. System matches face, checks liveness and document authenticity. 5. System screens AML and PEP lists. 6. System scores risk and approves with a credit limit. 7. Applicant and merchant are notified; everything is logged.	1. Analyst opens the case from the queue. 2. Analyst reviews evidence, scores and reason codes. 3. Analyst approves, rejects or requests more information and enters a reason. 4. System updates the status, notifies the applicant and logs the action.	1. System pauses the decision and creates an AML task. 2. Officer reviews the matching details against the applicant's data. 3. Officer marks it false positive or confirmed and records the justification. 4. Decision engine continues or the case is declined and logged.
Alternative and exception flows	Poor image: system asks for a retake (limited attempts). Face mismatch, failed liveness or low confidence: case flagged. Service failure: case queued for review and administrator alerted.	More information requested: applicant uploads again and flow returns to verification.	Confirmed match: no activation; record kept for the required period.
Postcondition	Applicant activated with a limit, or case flagged with reasons	Final decision recorded	Outcome and justification recorded

3.3 Functional Requirements

Priority uses MoSCoW: Must, Should, Could. Each requirement is linked to a problem from Section 1.1.

ID	Requirement	Problem	Priority
A. Intake and capture			
FR-01	The system shall start an onboarding session from a merchant QR code or the app link and record the merchant and channel.	P1	Must
FR-02	The system shall display a data-processing notice and record the applicant's explicit consent with time before any capture.	P5	Must
FR-03	The system shall let the applicant capture the front and back of the national ID, or a passport, and a selfie.	P1	Must
FR-04	The system shall check image quality (blur, glare, cropping, resolution) and ask for a retake with a reason, up to a configurable number of attempts.	P3	Must
B. OCR and data validation			
FR-05	The system shall extract name, national ID number, date of birth, address and expiry date from the ID using an AI OCR engine, in Arabic and English.	P3	Must
FR-06	The system shall provide a confidence score per extracted field and send low-confidence fields for review.	P3	Must
FR-07	The system shall validate the national ID number structure (14 digits, birth date, governorate code) and check consistency with the date of birth read from the ID.	P2, P3	Must
FR-08	The system shall detect expired documents and duplicate applications for the same ID number.	P2	Must
C. Identity verification			
FR-09	The system shall match the face on the ID with the selfie and return a similarity score.	P2	Must
FR-10	The system shall run a liveness check to reject static photos, screen replays and recorded video.	P2	Must
FR-11	The system shall check document authenticity (tampering, altered fonts or fields, digital manipulation, photocopy or screen capture) and return a score.	P2	Must
FR-12	The system shall treat borderline identity results as flagged cases, not automatic rejections.	P2, P4	Must
D. AML and PEP screening			
FR-13	The system shall screen each applicant against international and local sanctions, PEP and blocked-person lists as soon as data is extracted.	P5	Must
FR-14	The system shall use fuzzy matching that handles Arabic and English name variants and report the match strength.	P5	Must
FR-15	The system shall pause the decision on any possible match and create a task for the compliance officer.	P5	Must
FR-16	The system shall allow the screening lists to be updated and shall record the list version used for each screening.	P5	Should
E. Risk scoring and decision			
FR-17	The system shall compute a risk score from identity results, screening results and application data, and return reason codes.	P4	Must
FR-18	The system shall automatically approve applications that meet configured criteria and assign a credit limit by risk band.	P1, P4	Must
FR-19	The system shall flag applications that do not meet auto-approval criteria and route them to the review queue.	P4	Must
FR-20	The system shall let authorized users configure thresholds and credit limit rules without code changes.	P4	Should

ID	Requirement	Problem	Priority
FR-21	The system shall support a mock credit enquiry interface that can be replaced by a real credit bureau connection.	P3	Could
F. Review dashboard			
FR-22	The system shall show flagged cases in a queue with ID image, selfie, extracted data, scores, reason codes and screening results.	P4	Must
FR-23	The system shall let a reviewer approve, reject or request more information, and require a recorded reason.	P4	Must
FR-24	The system shall support role-based access (analyst, risk manager, compliance officer, administrator).	P5	Must
G. Notifications, audit and reporting			
FR-25	The system shall notify the applicant and the merchant of the result immediately through the same channel, with the next step if the case is under review.	P1	Must
FR-26	The system shall write an append-only audit log of every input, check result, decision and human action with time and actor.	P5	Must
FR-27	The system shall produce reports on volume, approval rate, review rate, rejection reasons and processing time.	P4	Should
FR-28	The system shall alert administrators on service failures and apply retries and timeouts, sending the case to review if a check cannot complete.	P4	Must
FR-29	The system shall delete or anonymize images and personal data when the retention period ends, while keeping the records required by law.	P5	Should

3.4 Non-functional Requirements

Category	ID	Requirement
Performance	NFR-01	A clean application shall reach a decision in about 3 minutes from QR scan, of which system processing shall take under 30 seconds.
Performance	NFR-02	Design budget for processing: OCR 5 s, face match and liveness 10 s, document authenticity 10 s, AML screening 5 s, scoring and decision 3 s, notification 2 s.
Performance	NFR-03	The system shall handle at least 20 concurrent onboarding sessions in the demo environment, and the design shall allow horizontal scaling.
Accuracy	NFR-04	OCR field-level accuracy shall be above 98% on the labelled test set; accuracy shall be reported per field and per image-quality level.
Accuracy	NFR-05	Face matching, liveness and authenticity checks shall be evaluated with measured false-accept and false-reject rates on the test set, and thresholds documented.
Security	NFR-06	All traffic shall use HTTPS; stored images and personal data shall be encrypted at rest.
Security	NFR-07	Access shall follow least privilege with role-based access control; credentials and API keys shall never be stored in the repository.
Security	NFR-08	The audit log shall be append-only and protected against modification; national ID numbers shall be masked in logs and notifications.
Privacy	NFR-09	Biometric and financial data shall be collected only with explicit consent, used only for onboarding, and deleted when the retention period ends [14].
Privacy	NFR-10	Personal data shall be processed and stored in Egypt where possible; any external AI service shall receive only synthetic data in the prototype [14].
Compliance	NFR-11	Screening and due-diligence records shall be retained for at least five years after the relationship ends, subject to confirmation with compliance [8].
Explainability and fairness	NFR-12	Every decision shall store reason codes; thresholds and results shall be reviewed for unfair differences between test groups; every rejection shall have a human-reviewed reason.
Usability	NFR-13	The customer flow shall be mobile-first, available in Arabic and English, completed without help by at least 90% of test users, and meet basic accessibility practices (contrast, text size).
Usability	NFR-14	Messages to applicants shall be short and say exactly what to do next (retake, wait for review, contact the shop).
Reliability	NFR-15	Service availability shall be at least 99% in the test period; failures shall not lose submitted data and shall fall back to manual review.
Maintainability	NFR-16	Each AI module shall sit behind a clear interface so that an OCR, face or screening provider can be replaced without rebuilding the flow.
Testability	NFR-17	A synthetic test dataset (genuine, low quality, tampered and spoof samples) shall be maintained in the repository documentation without any real personal data.

3.5 Data Requirements, Assumptions and Constraints

- **Data:** applicant details, ID images, selfie, extracted fields, verification scores, screening results, risk score and reason codes, decision, credit limit, reviewer actions, consent record and audit events.
- **Test data:** synthetic or fictional ID images and applicants only; no real national IDs, selfies or customer records are used in development, testing, the repository or the demo.
- **Lists:** the prototype uses public or sample sanctions and PEP data; a real deployment would use FlexiPay's licensed list provider.
- **Targets:** the 3-minute, 98% and 80/20 figures are design targets from the brief and will be reported honestly after testing.

- **Constraints:** delivery dates follow the Project Documentation Guidelines; open-source and free-tier tools; no production deployment; team of 5 members.

3.6 Traceability: problems, requirements and KPIs

Problem	Key requirements	KPIs (Section 1.5)
P1 Drop-off at point of sale	FR-01, FR-03, FR-04, FR-18, FR-25, NFR-01, NFR-13	Activation time, processing time, self-service completion
P2 Identity fraud	FR-07 to FR-12, NFR-05 (Planned for future production phase)	Forgery detection, spoof rejection, false rejection of genuine applicants
P3 Manual entry errors	FR-04 to FR-07, FR-21, NFR-04	OCR accuracy
P4 Operating cost	FR-12, FR-17 to FR-20, FR-22, FR-23, FR-27	Straight-through rate, review share, manual effort reduction
P5 KYC and AML compliance	FR-02, FR-13 to FR-16, FR-24, FR-26, FR-29, NFR-08, NFR-11 (Planned / Future compliance integration)	Screening coverage, audit completeness